

IT112/113 Introduction to Programming /Programming Lab

Semester I, 2025-2026

Lectures **TBA**

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Course Website **TBA**

Course Objective

This course aims to introduce problem solving techniques using programming to help the students to develop analytical and logical skills. The course starts with basic concepts of computer programming and follows in building up knowledge in program development, deployment and testing to solve computational problems. The course also provides visualization of memory and time requirements for solving problems using the C programming language. The coverage of this course includes problem solving techniques, flow charts, algorithms development, pseudo codes, and implementation of algorithms using C programming.

Course Topics include primitive data types, control structures, structured programming, arrays, strings, functions, and pointers including memory allocation and deallocation efficiently.

We will also cover some ways to constructively use concepts from and GenAI tools like LLMs to supplement programming. The lab course IT113 is closely coupled with this course.

Course Outcome

At the end of the course, students will be able to develop logical and analytical ability to perceive and solve computational problems; to write and test computer programs developed with C programming language; and to work effectively with various programming tools like editors, compilers, debuggers etc.

Program outcomes

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X	X		X			X	X			X

Evaluation Scheme

Mid Semester Test	40%
Final Examination	60%

Course Conduct and Grading Policy (For IT112 and IT113)

1. Students are supposed to be present for all lectures and lab sessions.
2. If the attendance falls below 70%, a student will receive an F grade. (An institute-wide detailed policy will be communicated separately)
3. **All lab assignments (IT113) must be completed**, failing which zero marks will be awarded for the lab course. One makeup lab session will be scheduled at the end of the semester for students who miss a session due to any reason. For special medical cases, if the Dean, AP has approved their application, additional makeup sessions may be provided to the concerned students.
4. If a case of plagiarism/copying is detected in the exams or lab submissions (IT112/ IT113), then that student will **receive zero marks** in the **complete lab segment** or the respective **exam component**. The student will also be reported to the Dean, AP for further actions.

IT112 Course Content

1. Core C (20 Hours)

Variables, Types, Assignment, Operators, Expressions, Branching and Looping
Solving problems by representing solutions as pseudocode. Mapping pseudocode to C programs. Arrays. Functions, passing by value. Scoping rules. Basic I/O and File I/O using standard C library.

2. Tools (4 Hours)

Editor, Compiler, Debugger, Make, and Shell commands. (We will be using Linux/gedit/gcc/gdb/make/bash for this course.) Multi-file programs.

3. Pointers (10 Hours)

Pointers. Pointer arithmetic. Function - pass by reference. Command line arguments. Dynamic memory allocation. Struct and link list.

4. Recursion (4 Hours)

Basic concepts. Solving simple problems using recursion. Performance comparison with iterative solutions.

Lab Content and Exercises will be tightly tied with content covered in the class.

Textbook

Yashvant Kanetkar, **Let Us C**, 19th Ed, BPB Publications

References

1. Kernighan Brian, Ritchi Dennis, The C Programming Language (ANSI C Version), 2nd Ed., Prentice Hall
2. Deitel Harvey, C: How to Program, PHI Learning

Online compiler used in the lectures:

<https://onecompiler.com/c>

Other tools will be communicated to the students whenever they are used.